```
!pip install spacy networkx
!python -m spacy download en_core_web_sm
```

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✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy
import networkx as nx

nlp = spacy.load("en_core_web_sm")

def build_dependency_graph(doc):
    """Builds a NetworkX graph from a spaCy Doc's dependency parse tree."""
    edges = []
    for token in doc:
        for child in token.children:
            # Add edges between parent (token) and child
            edges.append((token.i, child.i, {"dep": child.dep_, "dir": "down"}))
    return nx.Graph(edges)

def get_shortest_dependency_path(doc, ent1_idx, ent2_idx):
    """Finds the shortest path between two token indices in the dependency graph."""
    graph = build_dependency_graph(doc)
    try:
        path_indices = nx.shortest_path(graph, source=ent1_idx, target=ent2_idx)
        # Reconstruct path details
        path_tokens = [doc[i] for i in path_indices]
        return path_tokens, path_indices
    except nx.NetworkXNoPath:
        return None, None

def extract_relations(text):
```

```
    doc = nlp(text)
    relations = []

    # Identify entity pairs
    entities = list(doc.ents)
    num_entities = len(entities)

    print(f"Detected Entities: {[(e.text, e.label_) for e in entities]}\n")

    for i in range(num_entities):
        for j in range(i + 1, num_entities):
            ent1 = entities[i]
            ent2 = entities[j]

            # We use the root token of each entity phrase for finding the dependency path
            ent1_root = ent1.root
            ent2_root = ent2.root

            path_tokens, path_indices = get_shortest_dependency_path(doc, ent1_root.i, ent2_root.i)

            if path_tokens:
                # Build subsequence representation: e.g. Entity1 -> dep_path -> Entity2
                path_str = " -> ".join([f"{t.text} ({t.dep_})" for t in path_tokens])
                relations.append({
                    "Entity1": ent1.text,
                    "Entity1_Type": ent1.label_,
                    "Entity2": ent2.text,
                    "Entity2_Type": ent2.label_,
                    "Shortest_Dependency_Path": path_str,
                    "Path_Tokens": [t.text for t in path_tokens]
                })
    return relations
```

```
sample_text = "Google was founded by Larry Page and Sergey Brin in California."
relations = extract_relations(sample_text)

import pandas as pd
df_relations = pd.DataFrame(relations)
display(df_relations)
```

```
Detected Entities: [('Google', 'ORG'), ('Larry Page', 'PERSON'), ('Sergey Brin', 'PERSON'), ('California', 'GPE')]
```

	Entity1	Entity1_Type	Entity2	Entity2_Type	Shortest_Dependency_Path	Path_Tokens
0	Google	ORG	Larry Page	PERSON	Google (nsubjpass) -> founded (ROOT) -> by (ag...	[Google, founded, by, Page]
1	Google	ORG	Sergey Brin	PERSON	Google (nsubjpass) -> founded (ROOT) -> by (ag...	[Google, founded, by, Page, Brin]
2	Google	ORG	California	GPE	Google (nsubjpass) -> founded (ROOT) -> in (pr...	[Google, founded, in, California]
3	Larry Page	PERSON	Sergey Brin	PERSON	Page (pobj) -> Brin (conj)	[Page, Brin]